Assignment 1

CMPS 260: Introduction to Linux – January 2026

Introduction

Linux comes in many different distributions, each designed for different users and purposes. Choosing the right distribution is important because it affects how easy it is to set up, manage, and complete a project. In this report, I review several Linux distributions and determine which one is the best fit for my final project.

Linux Distributions Researched

Ubuntu

Target Audience:
Beginners, students, developers, and system administrators.

Key Features and Benefits:
Ubuntu is easy to install and widely used. It has Long Term Support (LTS) versions that are stable and reliable. Ubuntu supports most development tools out of the box, making it a strong choice for backend and server projects.

Challenges or Limitations:
Ubuntu uses slightly more system resources than minimal distributions, but this is usually not an issue for small projects.

Community Support and Documentation:
Ubuntu has a very large community, with plenty of tutorials, forums, and documentation available online.

Debian

Target Audience:
System administrators and users who prioritize stability.

Key Features and Benefits:

Debian is known for being extremely stable and efficient. It is often used on servers that require long-term reliability.

Challenges or Limitations:

Debian's software packages can be older, which may slow development. It can also be more difficult for beginners to configure.

Community Support and Documentation:

Debian has strong documentation, but the community is more technical and less beginner-focused.

Fedora

Target Audience:

Developers and advanced Linux users.

Key Features and Benefits:

Fedora includes newer Linux features and tools. It has good security support and is often used for testing modern technologies.

Challenges or Limitations:

Fedora has a short support lifecycle and requires frequent updates, which is not ideal for a short class project.

Community Support and Documentation:

Fedora has decent documentation, but a smaller community compared to Ubuntu.

Final Project Description

Project Title:

Linux Server Health Monitor

Project Overview:

The final project will be a simple backend system that monitors basic Linux server information such as CPU usage, memory usage, disk space, and system uptime. The data will be collected using standard Linux commands and made available through a basic API.

Project Requirements:

- Stable and reliable Linux environment
- Easy setup and maintenance
- Support for Python and Flask
- Ability to run Linux system commands
- Good documentation for troubleshooting

Distribution Comparison for the Project

Ubuntu best meets the project requirements because it is easy to use, stable, and compatible with Python-based backend tools. Debian is very stable, but may have slow development due to older packages. Fedora offers newer features but requires frequent updates, which could add unnecessary complexity.

Final Recommendation

Based on the comparison, Ubuntu is the best Linux distribution for the Linux Server Health Monitor project. It provides the right balance of simplicity, stability, and community support, allowing the project to focus on learning Linux infrastructure concepts rather than system configuration issues.

Conclusion

Different Linux distributions serve different needs. For this project, Ubuntu is the most practical choice because it supports fast development, reliable performance, and strong learning outcomes for an introductory Linux course.